ALKHIDMAT SUMMER INTERNSHIP 2026 | GEN Z ENTREPRENEURSHIP INITIATIVE

THANDA GODAM

Village cold storage, rented by the crate.

A small grower with 200 kg of tomatoes has nowhere to keep them and no way to wait for a better price. We put a solar powered cold room inside the village and rent it out one crate at a time.

40%
of what growers harvest is lost after harvest

<1m
tonnes of cold storage against 13m tonnes grown

64%
of Pakistan's farms are under 5 acres

15 and 16 August 2026

The crop is grown. It is lost after harvest.

40%

of Pakistan's perishable harvest is lost between the field and the buyer

$1bn+

the value the Asian Development Bank puts on that loss every year

900,000

tonnes of cold storage capacity against 13 to 15 million tonnes grown

Where the loss actually happens

5-8%

at harvesting

15-20%

in handling the picked crop

10-12%

in transport to market

Almost all of it happens after the hard work of growing is already done.

Sources: Pakistan Today and FreshPlaza on the 40 percent figure, Dawn reporting on the ADB estimate, AgriHunt on stage by stage losses. Full list at the end.

02 WHY IT HAPPENS

A small grower has about 48 hours to sell.

This is the part that most storage projects miss.

1 Day 0

He picks 200 to 400 kg. There is no cool place on the farm. The crates sit in the shade of a tree.

2 Day 1

Heat starts working on the crop. Tomatoes soften, leafy greens wilt, weight drops.

3 Day 2

It has to move now or it is worth nothing. He loads it and takes whatever the mandi pays that morning.

Small growers do not lose money because they farm badly.
They lose it because they cannot wait.

Tomato went from roughly PKR 90 a kg to PKR 400 and above within a few weeks this year. Everyone in the chain who can wait captures that movement. The grower, who cannot, ends up with 20 to 30 percent of what the customer finally pays.

03 THE GAP

Cold storage exists. Just not for a two acre farm.

Pakistan has around 555 cold storage units. Not one of them will take ten crates for ten days.

	What the market offers today	What a small grower needs
Size	1,000 tonnes and up	**5 tonnes, in the village**
Location	Industrial areas near big cities	**Within 15 minutes of the field**
Booking	Season long contracts	**A few days at a time**
Minimum	Full truckloads	**Ten crates**
Power	Grid, so load shedding hits the stock	**Solar, so it keeps running**
Customer	Traders and large commission agents	**The grower himself**

The gap is not only how much capacity exists. It is who the capacity was built for.

What the evidence already says.

And the part that still has to be asked in the field.

Desk research

- Pakistan Bureau of Statistics, 7th Agricultural Census: 64 percent of farms are under 5 acres, 26 percent under one acre, average farm size now 5.1 acres.
- Asian Development Bank, reported in Dawn: post harvest losses cost Pakistan over one billion dollars a year.
- Trade press on the storage gap: about 555 cold storage units holding roughly 900,000 tonnes against 13 to 15 million tonnes of fruit and vegetable production.
- IGC study on the fresh produce supply chain, and PIDE's counter argument that the commission agent also supplies credit. We took that seriously and built around him rather than against him.

The three questions that decide it

From published evidence. Growers interviewed next.

1. **How long can he hold the crop?**

   Days at Pakistani field temperatures. Trials on a Punjab tomato variety hold it for weeks at 10 degrees and 90 percent humidity.

2. **Is PKR 12 per crate per day payable?**

   Big cold stores charge PKR 3 to 6 per kg a month. We cost more per day and are only paid for the days used.

3. **How much does he actually lose?**

   15 to 20 percent goes in handling the picked crop, before the 10 to 12 percent lost in transport.

Sources for each of the three answers are listed on the final slide.

05 THE SOLUTION

Thanda Godam

A 5 tonne solar powered cold room, placed inside the village, rented by the crate by the day.

1

Close

Within 15 minutes of the field, so the crop goes in cold the same morning it is picked.

2

Small

Priced per 20 kg crate per day. No deposit, no season contract, no truckload minimum.

3

Independent

Runs on solar with battery backup, so load shedding does not spoil what is inside.

We are not selling cold storage.

We are selling the ability to wait.

06 HOW IT WORKS

Five steps, all of them already familiar.

1

Bring

He brings crates the same day he picks. Weighed, tagged and photographed at the gate. He keeps a printed receipt.

2

Pay

PKR 12 per 20 kg crate per day, in cash or on easypaisa. He pays for the days he uses and nothing else.

3

Store

Held at the right temperature and humidity for that crop. Tomato, onion, potato, chili and leafy greens each have their own setting.

4

Choose

He takes his crates out whenever he wants and sells them himself at the mandi. Nothing is locked in.

5

Or sell through us

If he prefers, we grade, pack and move his crop to city buyers in one aggregated load and keep a 7 percent service margin.

Steps 1 to 4 are the whole service. Step 5 is optional for the grower and is where most of our income comes from.

07 TARGET AUDIENCE

Who uses it, who benefits, and who pays.

Who uses it

- Growers under 5 acres, working tomato, chili and leafy vegetables first. These are the crops whose prices move week to week.
- They harvest in small lots, sell within two days, and have no storage of their own.

Who pays

- The grower pays the crate fee, in small amounts he can see the return on.
- City buyers pay our 7 percent service margin on aggregated, graded loads.

Who else gains

- Households eat better because income steadies across the season.
- City consumers see a slightly steadier supply and less waste in the chain.

How many people that is

64%
of all farms in Pakistan are under 5 acres

26%
are under a single acre

300-400
growing households within 6 km of the pilot site, our own estimate

Pilot site: a union council in Depalpur tehsil, Okara district. Cold storage there was built for potato traders. The vegetable growers alongside them have none.

The grower's arithmetic on one load of tomatoes.

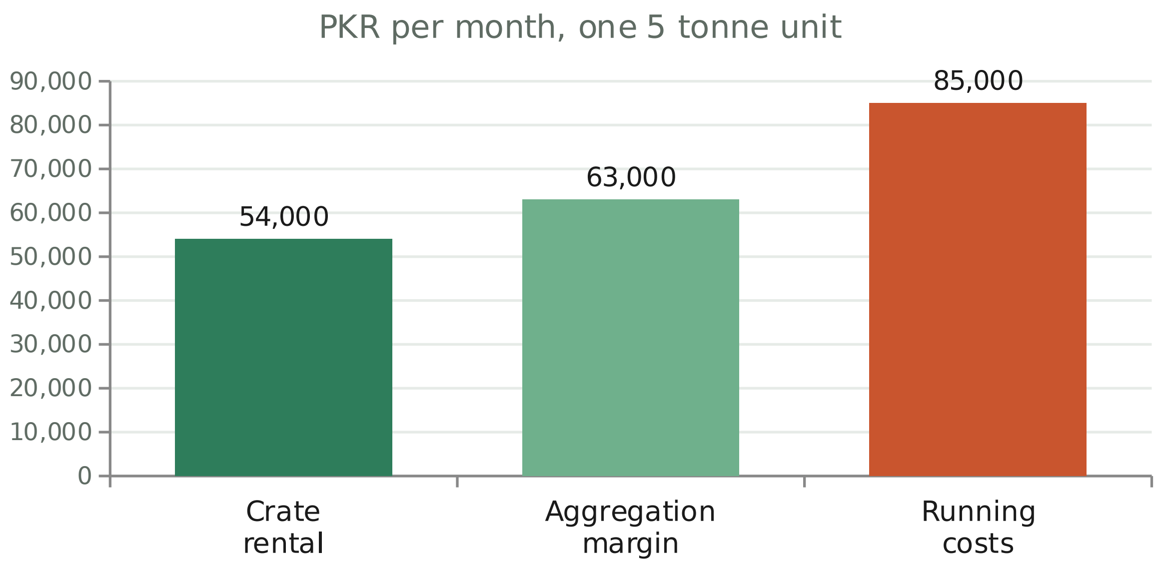
200 kg picked, ten crates, held for ten days.

What he pays	10 crates x PKR 12 x 10 days	**PKR 1,200**
What he saves	25 percent of 200 kg would have spoiled, so 50 kg kept	**50 kg**
What that is worth	50 kg at a PKR 60 per kg farm gate price	**PKR 3,000**
What he still has left	the choice to sell on a better day instead of a bad one	**upside on top**

About PKR 2.5 comes back for every PKR 1 he spends on storage, before he has sold anything at a better price.

The PKR 60 per kg farm gate price is a conservative in season figure. Both it and the spoilage rate move with the crop and the week.

One unit, two income lines, per month.



Crate rental — PKR 54,000

250 crate capacity at 60 percent average occupancy, PKR 12 a day

Aggregation margin — PKR 63,000

15 tonnes handled a month, 7 percent on a PKR 60 per kg value

Running costs — PKR 85,000

operator 30,000, loader 20,000, transport 15,000, maintenance 8,000, packaging 7,000, phone and misc 5,000

PKR 32,000

surplus a month, once the room is running

How it is funded, honestly

At an estimated PKR 2.5 million to build, one unit repays itself in about six and a half years. That is too slow for an equity investor and normal for zakat, CSR or green financing capital. So we raise the building cost once as a grant, and the unit covers its own running costs from the first season and helps fund the next one.

Four things already serve this farmer.

None of them do what we are proposing to do.

Commercial cold storages

Around 555 units nationally. Built for traders on season contracts and truckload minimums, and sited near cities.

Cannot take ten crates for ten days.

Agritech marketplaces

Tazah, Bazaar and others matched buyers and sellers digitally. Tazah has since pivoted away from local agri trade twice.

They move produce. They do not fix where it waits.

Government and donor schemes

Cold chain grants and subsidised storage funds exist and are useful.

The capacity added is small next to a 40 percent loss.

The commission agent

He advances credit before sowing and guarantees offtake at harvest. That is real value, and PIDE is right that he is not simply a villain.

We do not replace him. We rent him crate space at the same rate.

The only storage in the chain priced by the crate, by the day, and sited where the crop is picked.

Village marketing is not digital ads.

1 Friday announcement

A two minute announcement after Friday prayers in three village mosques, made by someone the village already knows.

2 The input dealer's shop

Every grower visits the seed and fertiliser shop. A small stall and a signboard there reaches more people than any advertisement.

3 Free first season

The first 20 growers store free for one season in exchange for letting us record their weights, prices and dates.

4 A WhatsApp rate group

Daily mandi rates sent to every grower who signs up. It costs nothing and gives them a reason to stay in contact.

5 Alkhidmat's district network

Existing offices and volunteers make the introductions and carry the trust we have not earned yet.

6 A board at the gate

What growers who waited actually earned, written up where everyone can see it. Proof travels faster than promises.

The first season is not about volume. It is about twenty growers who can tell the next hundred what happened to their money.

12 RISKS

What could go wrong, and what we would do.

1	**Growers do not trust us with their crop**	Weigh, tag and photograph at the gate. Printed receipt every time. A village committee with two growers on it holds the second key.
2	**Arguments over quality on collection**	The grade is agreed and written on the receipt when the crop goes in, with the entry photograph attached to it.
3	**The whole market crashes, not just one day's price**	Storage buys days, it does not buy a different market. We lead with tomato, chili and leafy greens, where prices move week to week.
4	**The room sits empty out of season**	Rotate to crops with different calendars, and offer the same room for seed, dairy and mango season overflow.
5	**Solar or compressor breakdown**	A service contract with the supplier from day one, spare parts held locally, and a small generator that can run the compressor alone.
6	**The commission agent works against us**	He is offered crate space at the same rate as everyone else. We compete with his storage, not with his credit.
7	**Capital does not arrive**	We can start with a 2 tonne room at a much lower cost and grow from the surplus, which delays the plan but does not end it.

13 NEXT STEPS

The next 90 days

Nothing here needs the full capital. All of it can start this month.

1

Weeks 1 to 3

Field validation

12 grower interviews, 3 commission agents, 2 storage operators. Settle the one question that matters: will he pay PKR 12 a crate.

2

Weeks 4 to 6

Real costs

Three written quotes for a 5 tonne solar cold room. Replace our PKR 2.5 million estimate with a number from a supplier.

3

Weeks 7 to 9

Site and buyers

A written site agreement with one village, support from the union council, and a letter of intent from two city buyers.

4

Weeks 10 to 12

Capital and build

Put the case to Alkhidmat, a CSR partner and one green financing window. Begin the build on the pilot unit.

WHAT WE ARE ASKING FOR

Capital for one pilot unit, a village willing to host it, and an introduction to two city buyers. One room, one season, and a real number at the end of it.

APPENDIX

Sources

- Pakistan Bureau of Statistics, 7th Agricultural Census, reported by Profit and The Express Tribune, August 2025. Farm size distribution.
- Asian Development Bank estimate on post harvest losses, reported in Dawn.
- Pakistan Today and FreshPlaza, 2026. Forty percent post harvest losses and the cold storage gap.
- AgriHunt. Stage by stage breakdown of losses in fruits and vegetables.
- The Express Tribune, 2026. Storage fund set against farm losses.
- Dawn, 2026. Punjab potato growers face collapse as oversupply deepens. Prices of PKR 20 to 25 a kg and per acre losses.
- Pakistan Today, August 2026. Tomato moving from roughly PKR 90 to PKR 400 and above a kg.
- Pakistan Horticulture Development and Export Company, potato challenges and prospects in Okara.
- International Growth Centre, PAK-22066, March 2023. Understanding the fresh produce supply chain.
- PIDE. The role of the middleman and neglected aspects. Arab News and Dawn on the arhti system.
- Post harvest trials on Punjab tomato varieties comparing ambient storage with 10 degrees at 90 to 95 percent humidity.

Figures that are ours rather than published, and are marked as such on the slides: the PKR 2.5 million build cost, the 60 percent occupancy assumption, the 7 percent aggregation margin, the PKR 12 crate price, and the 300 to 400 household catchment.